Performance Review

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Department: R&D

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Developing a mathematical model for combined braking system (CBS) mechanism of motorcycles & optimizing the linkage design to meet IS 14664 regulation with 20% performance buffer.

Introduction

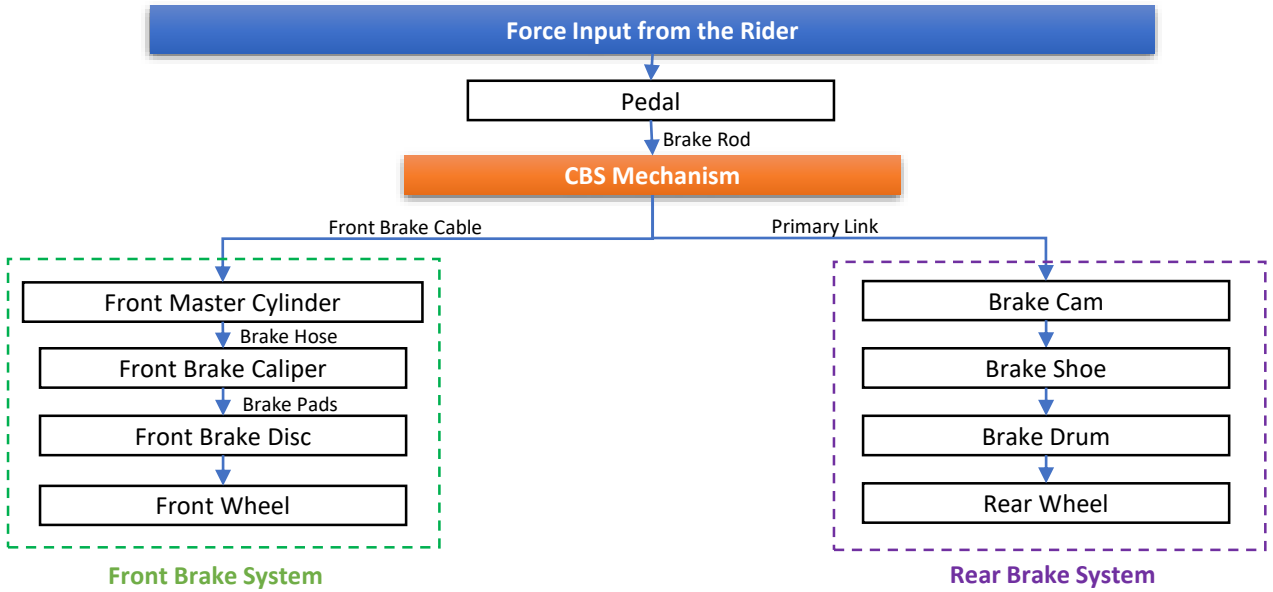
Scope: Covers regulatory background, mechanical design, force distributions, why a variable ratio is beneficial and the advantages of variable brake force ratio in CBS.

Methodology

1. **Literature Review-** Studied IS 14664 requirements for motorcycle braking systems.
2. **Regulatory Analysis-** Analyzed existing CBS mechanisms and identified performance gaps.
3. **Mathematical Modelling-** Developed equations to simulate braking force distribution between front and rear wheels.
4. **Leverage Optimization-** Modified linkage dimensions (equalizer and secondary links) to achieve better performance.
5. **Advantages-** Ensured proposed design met or exceeded IS 14664 with a 20% performance buffer.

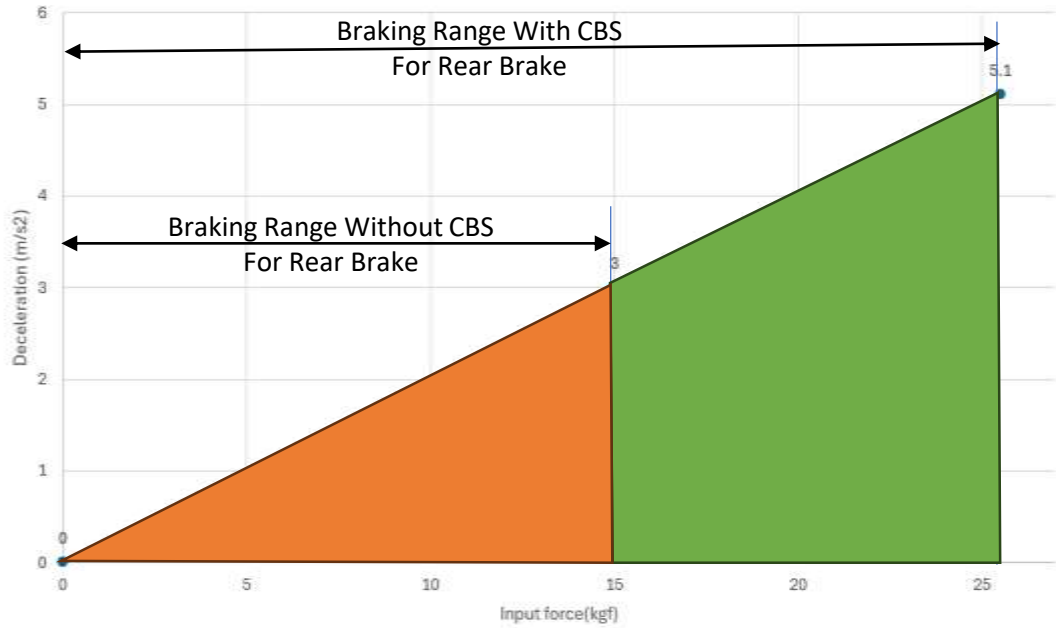
CBS Definition & Regulations as per IS14664: 2010

A **Combined Braking System** (CBS), is a safety feature in motorcycles and scooters that links the front and rear brakes.



CBS Mechanism Boundary Diagram (Disc-Drum configuration)

Rear Brake Input force (kgf) vs Deceleration (m/s²)



	Min. Deceleration Required
Rear Brake only (3-3)	2.9 m/s²
Rear Brake with CBS (3-3)	5.1 m/s²

Why People use Rear Brake over Front Brake? Even though Front is superior according to Braking Physics.
Indian riders often favor the rear brake for stability on crowded, low-grip roads. This preference stems from three main factors:

- **Road Conditions:** The chaotic and unpredictable nature of Indian traffic, with its constant need for minor speed adjustments, makes covering the rear brake a practical necessity.
- **Vehicle Dynamics:** Applying the rear brake gently settles the motorcycle's chassis without unsettling the steering, aiding stability and control during minor corrections.
- **Risk Perception:** Locking the rear wheel is generally perceived as more manageable and less dangerous than locking the front, which can lead to a complete loss of control.

Common Risk:
Excessive front braking- front wheel lock, front-flipping.
Ideal Braking:
Progressive front braking+ light rear pressure.

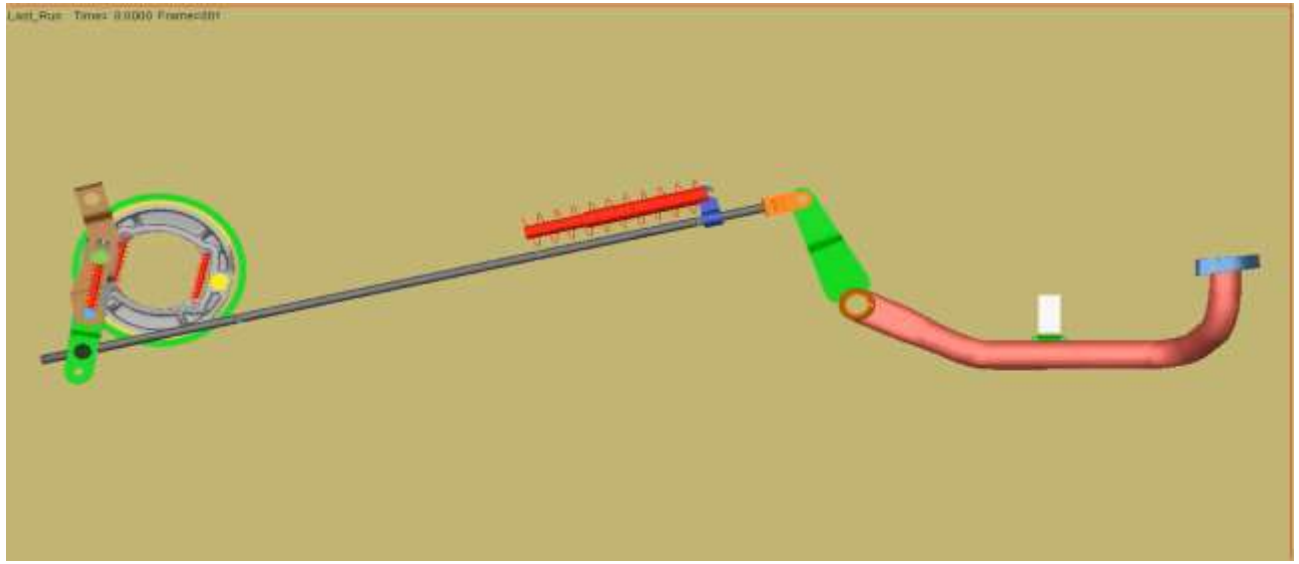
3-3 Vehicle: a two-wheeler with an engine above 50 cm³ or, regardless of propulsion type, a design speed over 50 km/h.

Existing CBS Mechanism

A **Combined Braking System** (CBS), is a safety feature in motorcycles and scooters that links the front and rear brakes.

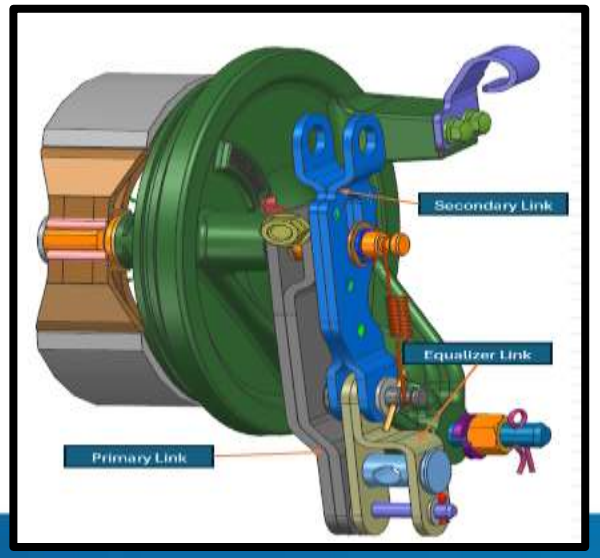
A force is applied on the brake pedal, as the pedal travel increases, the brake rod translates in its axis due to the **revolute joint**.

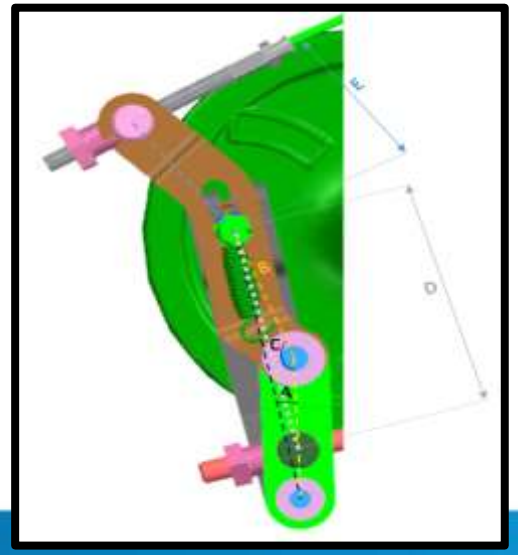
The other end of the brake rod is attached to the equalizer link



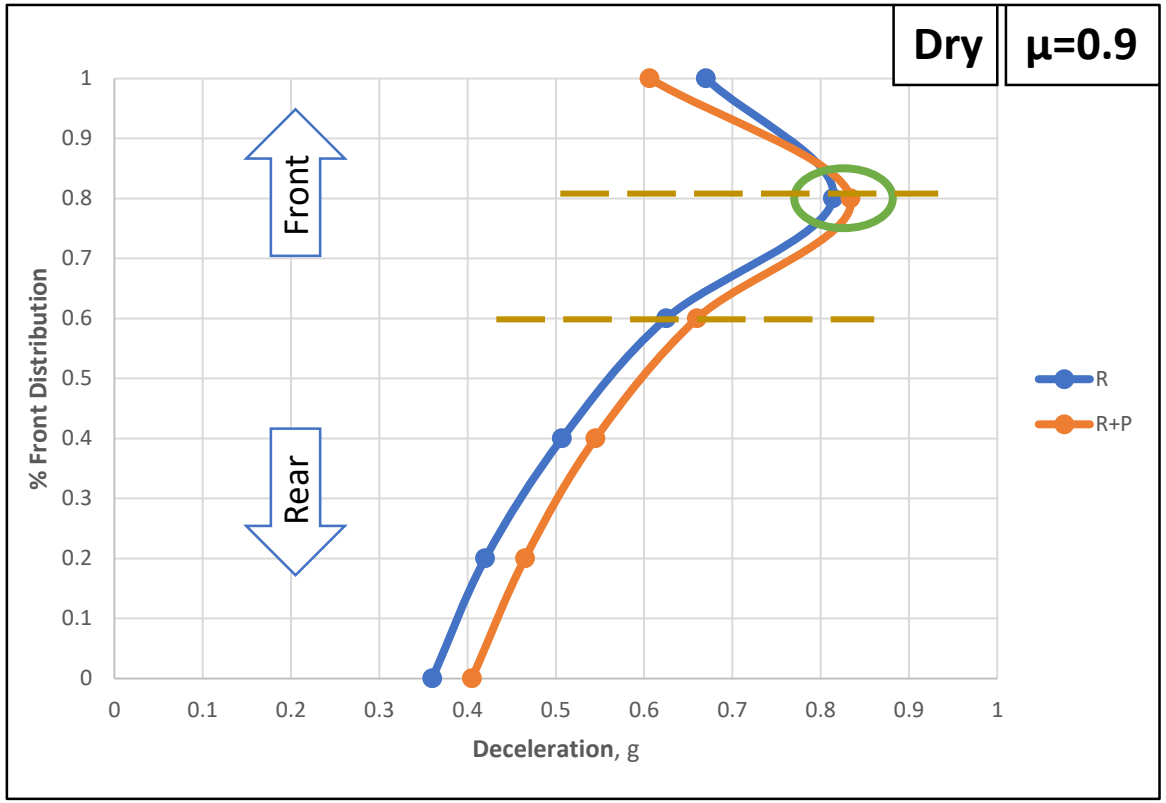
The CBS Mechanism:

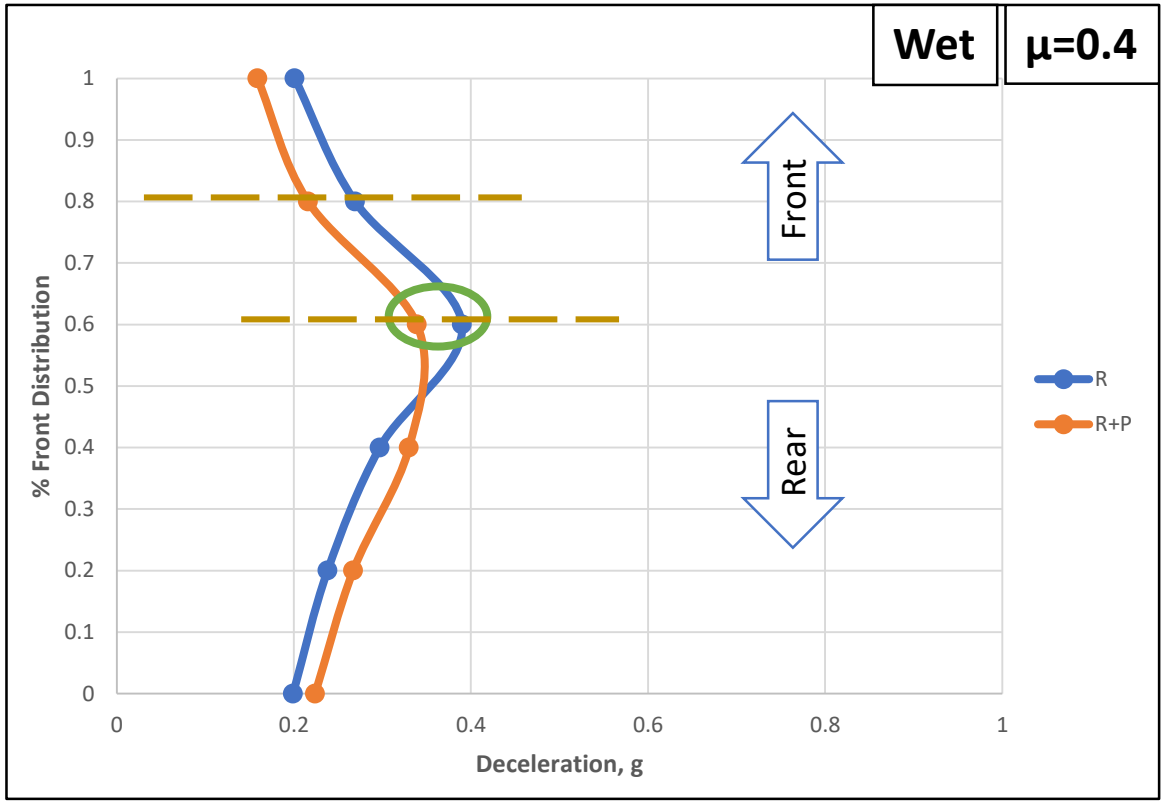
- The equalizer is basically divided into two parts, the lower end of the equalizer is attached to the primary link. The upper end is attached to the secondary link.
- The Cam revolutes about the panel, this actuates the rear mechanism. Secondary link moves along the slot, held in place with the help of return spring.





Highest Decelerations





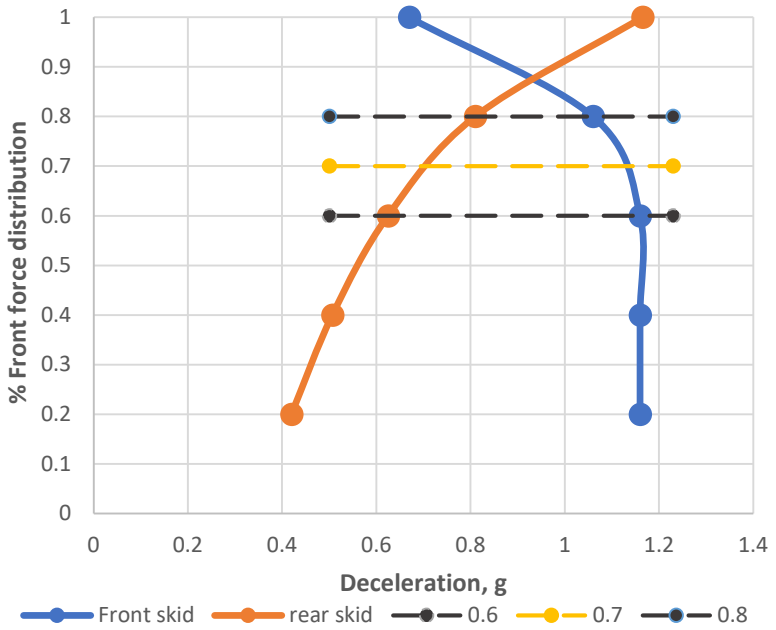
On **dry roads**, a higher front brake force is effective.

On **wet roads**, too much front force leads to early front skid and reduced performance.

A fixed force brake ratio results in premature skidding of the vehicle.

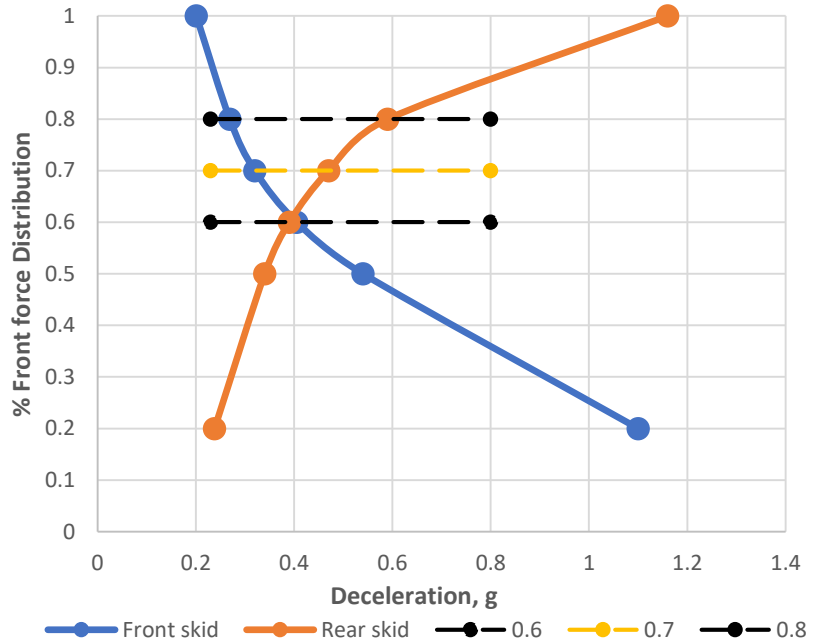
Skid Limit

Skid Limit for Dry road conditions (μ=0.9)

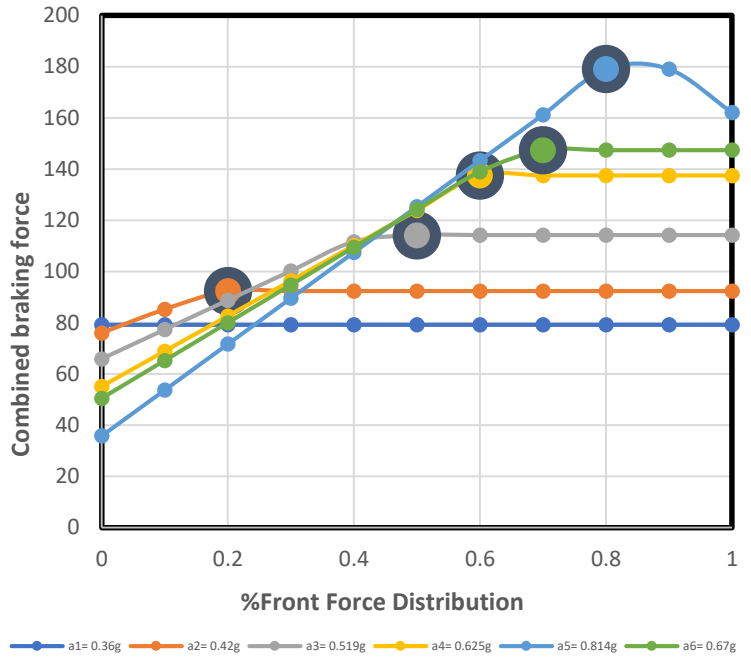


Beyond a certain ratio, the front wheel starts skidding before the rear.
This implies the front wheel is receiving more braking force than it can handle.

Skid Limit for Wet road conditions (μ=0.4)



Similar trend, but skids occur at much lower decelerations due to reduced friction.
Again, at low ratios, rear skids first — appropriate behavior.



The peak combined skid resistance varies with front- rear brake force distribution
So a fixed distribution does not support this.
A variable CBS is necessary to follow this peak for maximum braking efficiency without wheel lock.

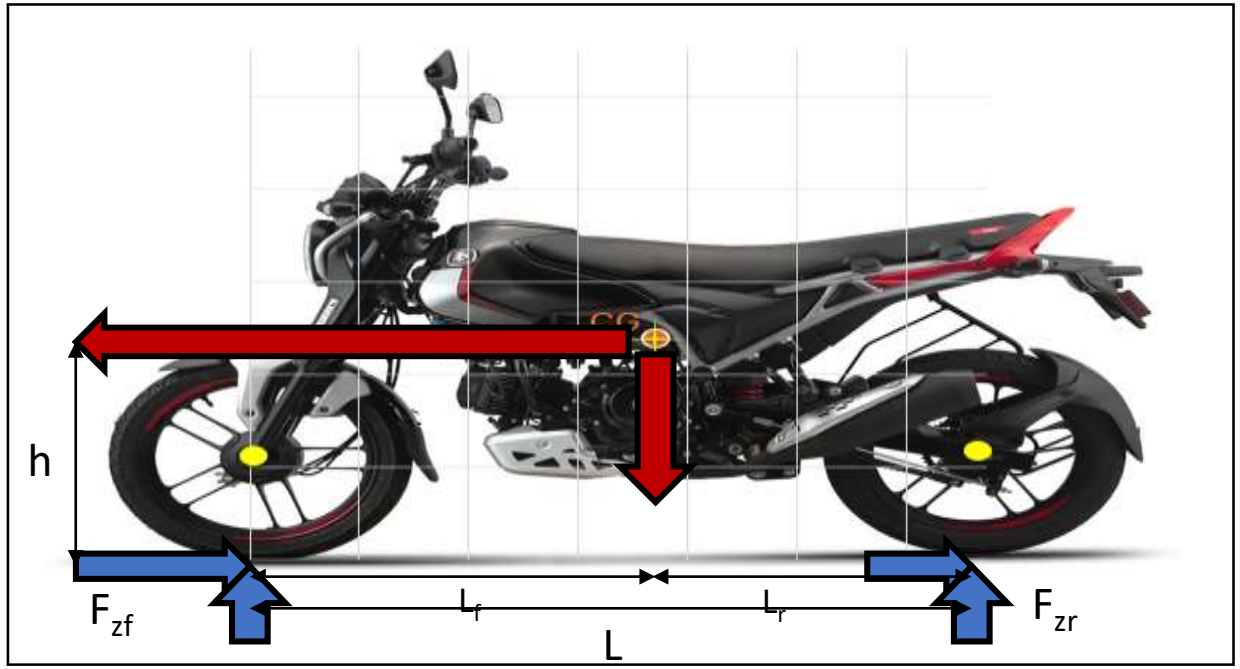
To maximize braking efficiency and prevent wheel lock under changing traction conditions, the system must **vary** the brake force ratio dynamically

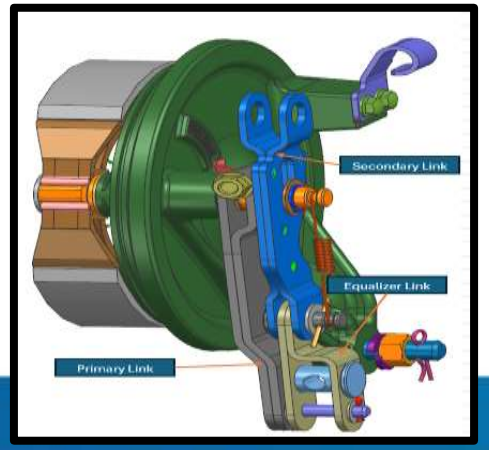
Build Breakdown- Freedom 125

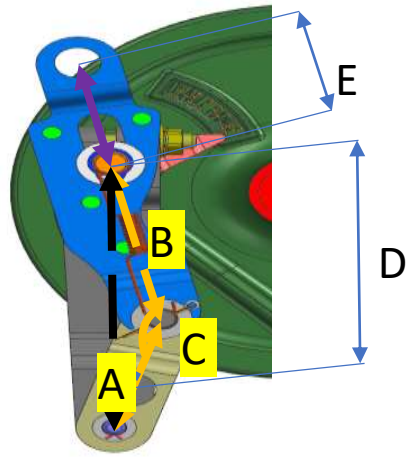
Parameter	Kerb	Rider
Wheelbase, mm (L)	1346	1343
FAW, Kg	65.25	88
RAW, Kg	79.75	132

Parameter	Kerb	Rider
Longitudinal location of CG from front axle, mm	740.3	804.6
CG height, mm (h)	478.2162	691.1015

	CBS link lengths for Braking Calculations	Existing
1	Pedal length, mm	249.10
2	Pedal output length, mm	86.00
3	Pedal leverage	2.90
4	Brake lever length, mm	108
5	Equalizer link length, mm	46.00
6	Secondary link total length, mm	104.00
7	Initial angle between brake lever and equalizer link, degrees (A)	0
8	Equalizer link length- Brake rod mtg point to secondary link mtg point, mm	30.00
9	Secondary link initial input length, mm (B)	62.00
10	Angle between s.no 9 & 10, degrees (C)	180.00
11	Length of virtual arm- brake rod mtg point to cam pivot, mm	92.00
12	Brake lever leverage input length, mm (D)	92.00
13	Secondary link output length, mm (E)	46.00







Correlation of B & L_{eq}

Let length of Brake lever be $\mathbf{L_{bl}}$

Let length of Equalizer link be $\mathbf{L_{eq}}$

After rear pedal free- play,

Let distance from brake rod mtg point to secondary link mtg point be $\mathbf{L_1}$

Let distance from brake rod mtg point to primary link mtg point be $\mathbf{L_2}$

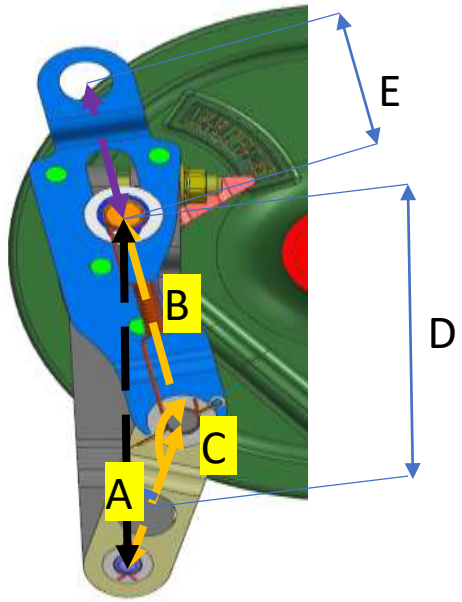
$A = \sin^{-1}(\text{Total brake rod pull at equalizer} / L_2)$

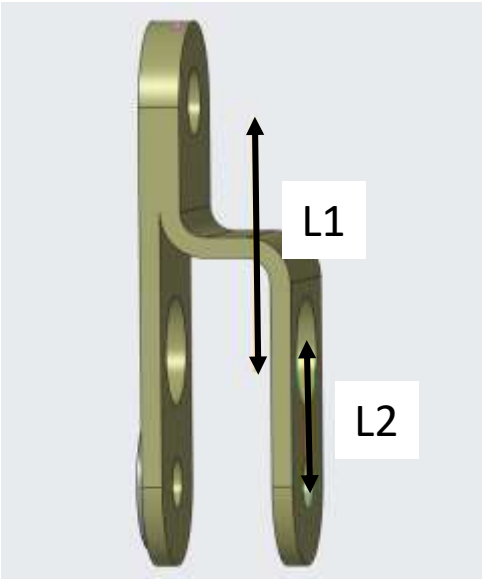
$B = \sqrt{[(L_{bl})^2 + (L_{eq})^2 + 2 * L_{bl} * L_{eq} * \cos(A)]}$

$C = \cos^{-1}[(B^2 + L_{eq}^2 - L_{bl}^2) / 2 * B * L_{eq}]$

$D = \sqrt{[L_1^2 + B^2 - 2 * L_1 * B * \cos(C)]}$

E= Total length of the secondary link - B





Front Leverage distribution ratio$\propto$ $\mathbf{L_{bl}/E}$

To increase front contribution Decrease E and maintain constant L_{bl}

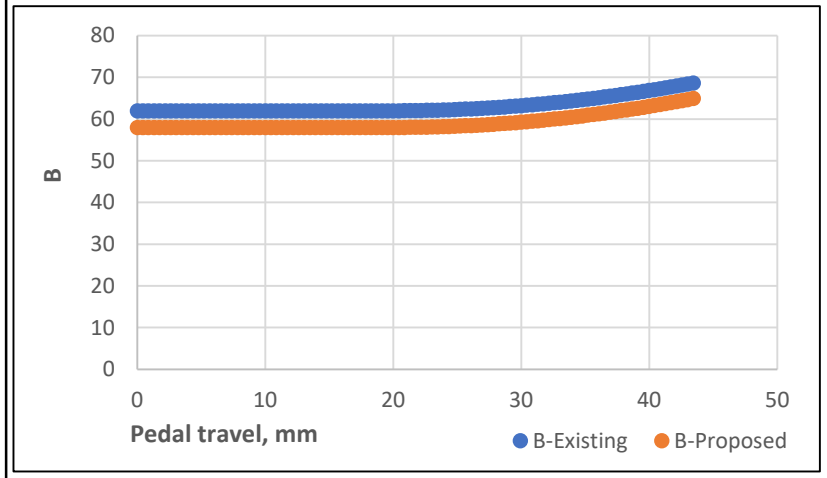
E depends on B, B has to increase so L_{eq} should also increase

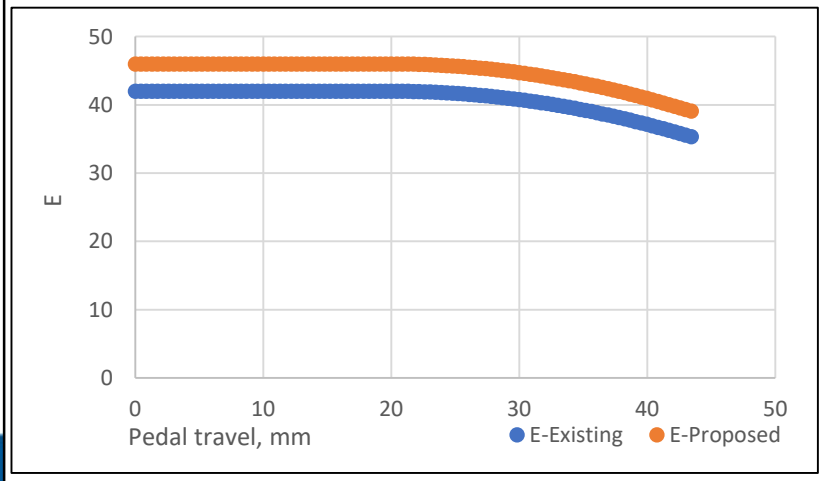
Leverage Optimisation

Correlation of B & L_{eq}

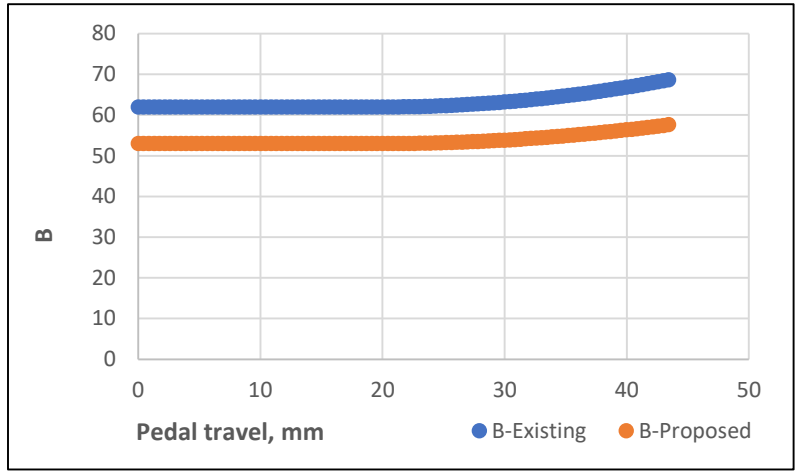
BAJAJ | THE WORLD'S FAVOURITE INDIAN

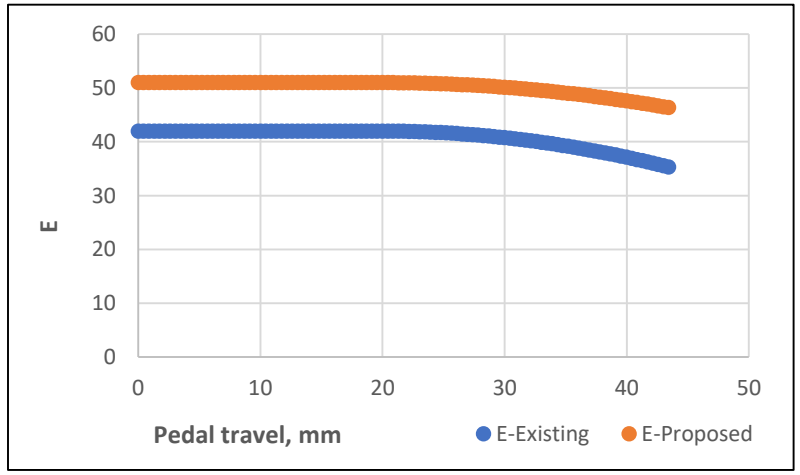
Case1: Decreasing Secondary link length by decreasing the length of Brake lever



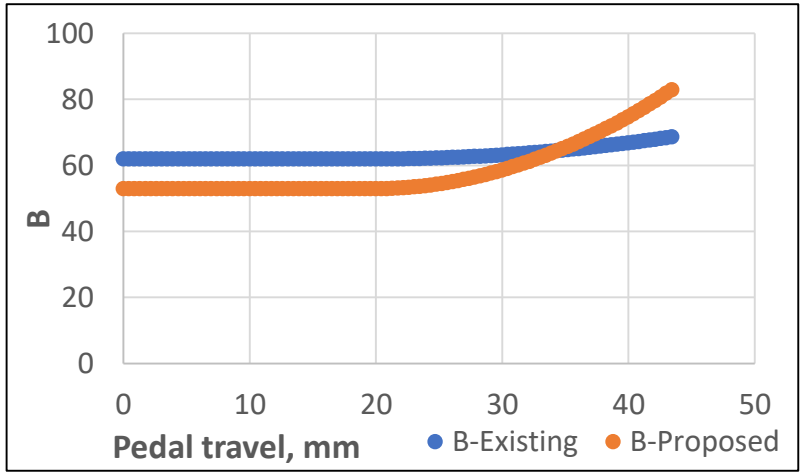


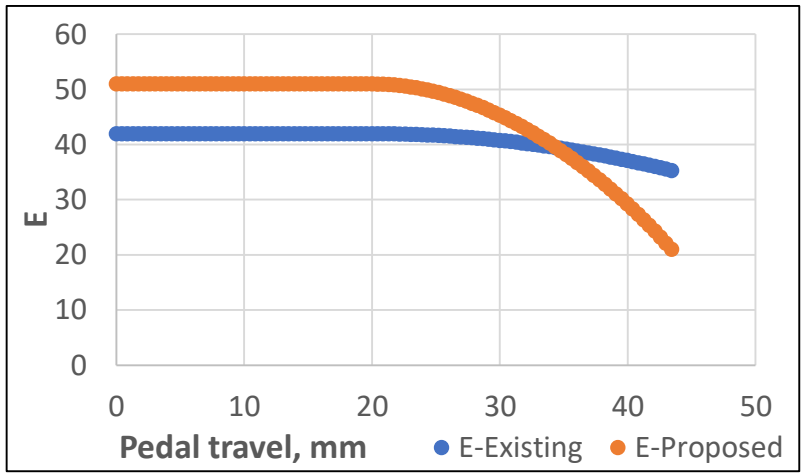
Case2: Increasing Equalizer link length by increasing distance from brake rod mtg to Primary link mtg





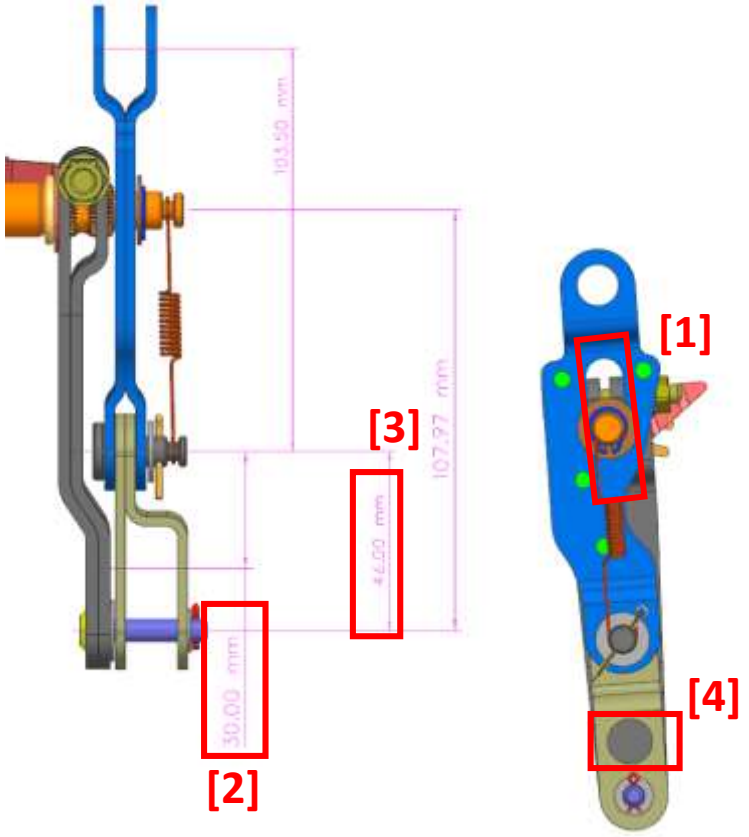
Case3: Increasing Equalizer link length by increasing distance from Brake rod mtg point to Secondary link mtg point



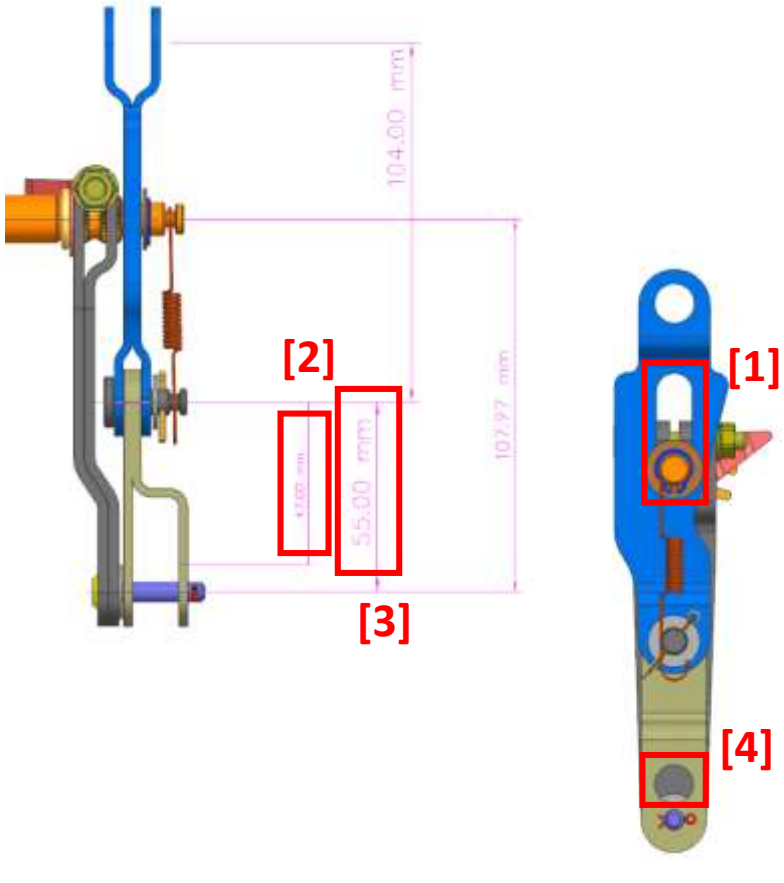


Existing vs Proposed Configuration

Existing Configuration



Proposed Configuration

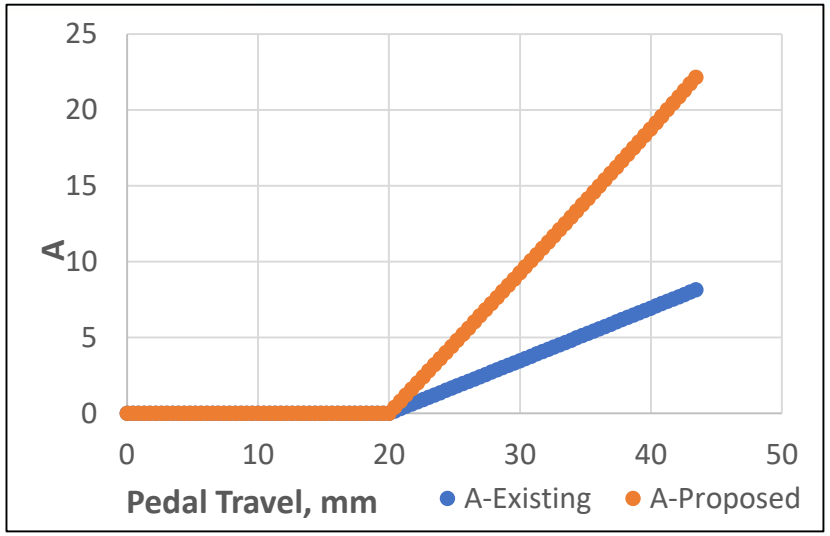


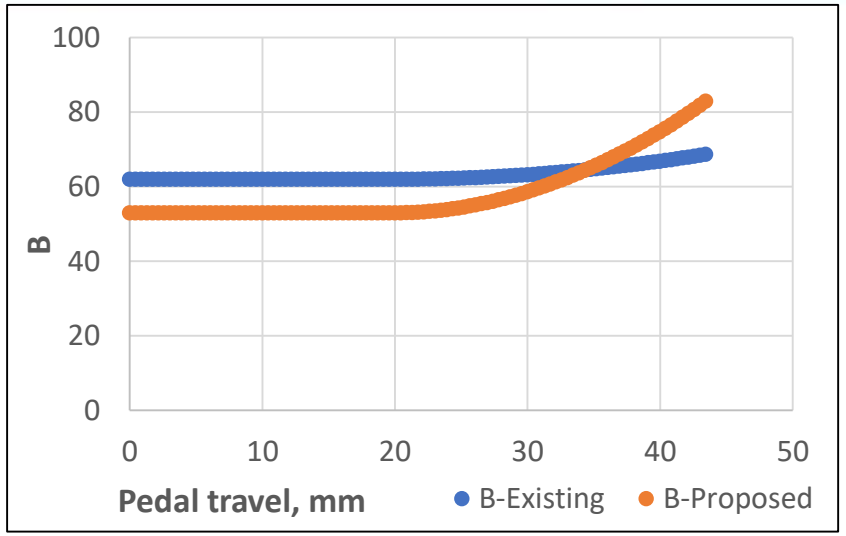
Design Changes:

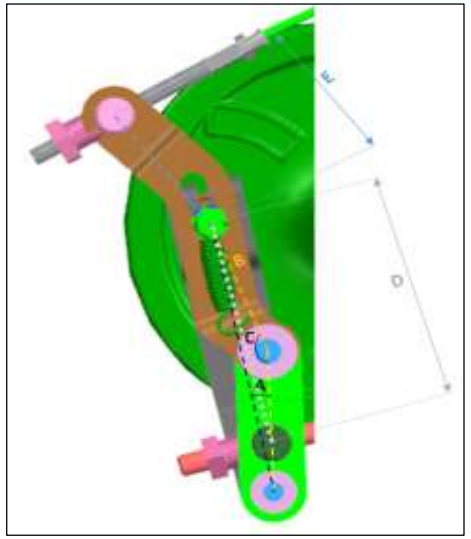
1. Slot Extended
2. Equalizer link length increased
3. Distance from Pivot to Cable Pin Hole
4. Brake Rod Cable Pin Hole diameter decreased

Parameter	Existing	Proposed
Brake lever length, mm	108.00	108.00
Equalizer link length, mm	46.00	55.00
Secondary link total length, mm	104.00	104.00
Initial angle between brake lever and equalizer link, degrees (A)	0	0
Equalizer link length- Brake rod mtg point to secondary link mtg point, mm	30.00	47.00
Secondary link initial input length, mm (B)	62.00	53.00
Angle between s.no 9 & 10, degrees (C)	180.00	180.00
Length of virtual arm- brake rod mtg point to cam pivot, mm	92.00	100.00
Brake lever leverage input length, mm (D)	92.00	100.00
Secondary link output length, mm (E)	46.00	51.00

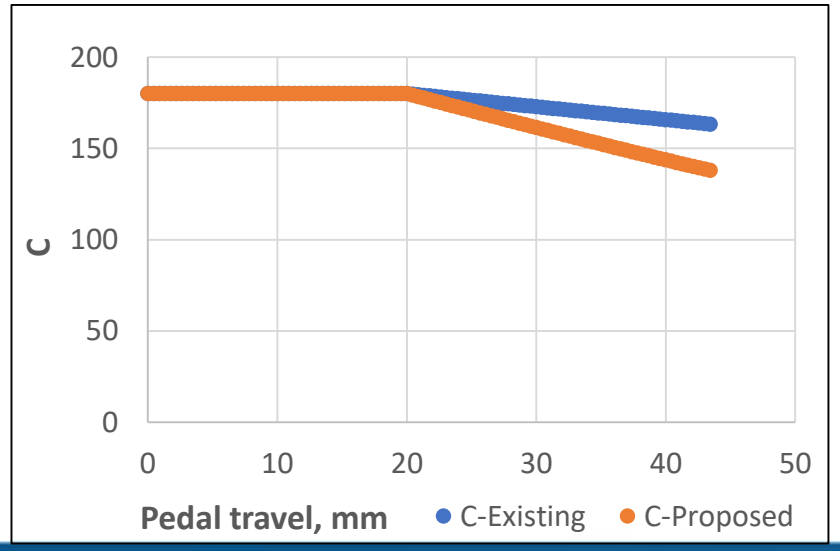
How Variability is Achieved

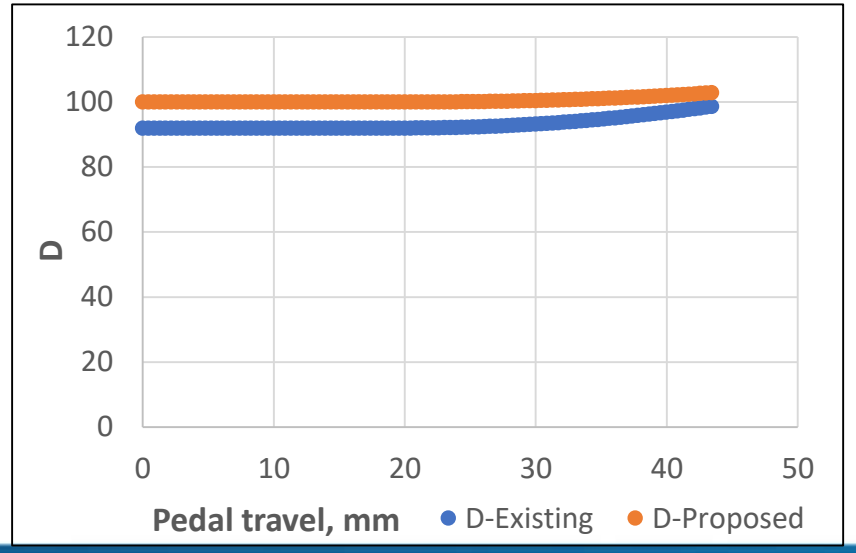


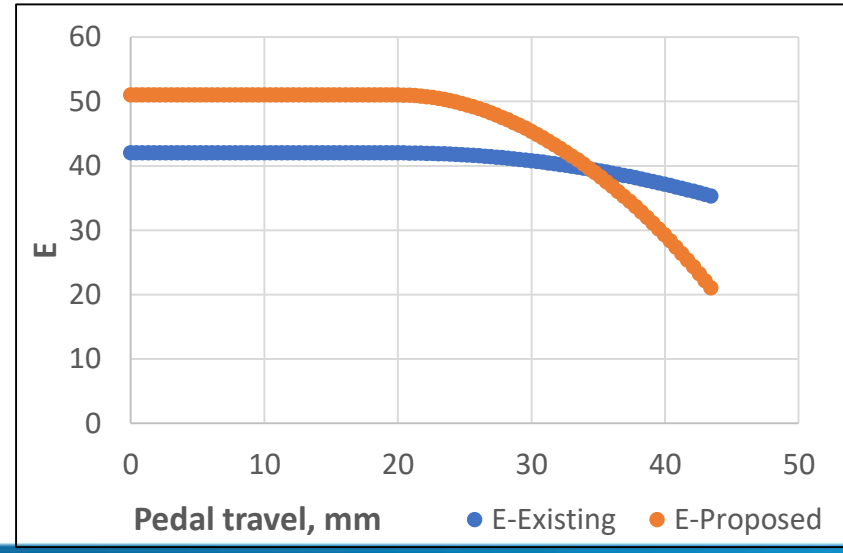




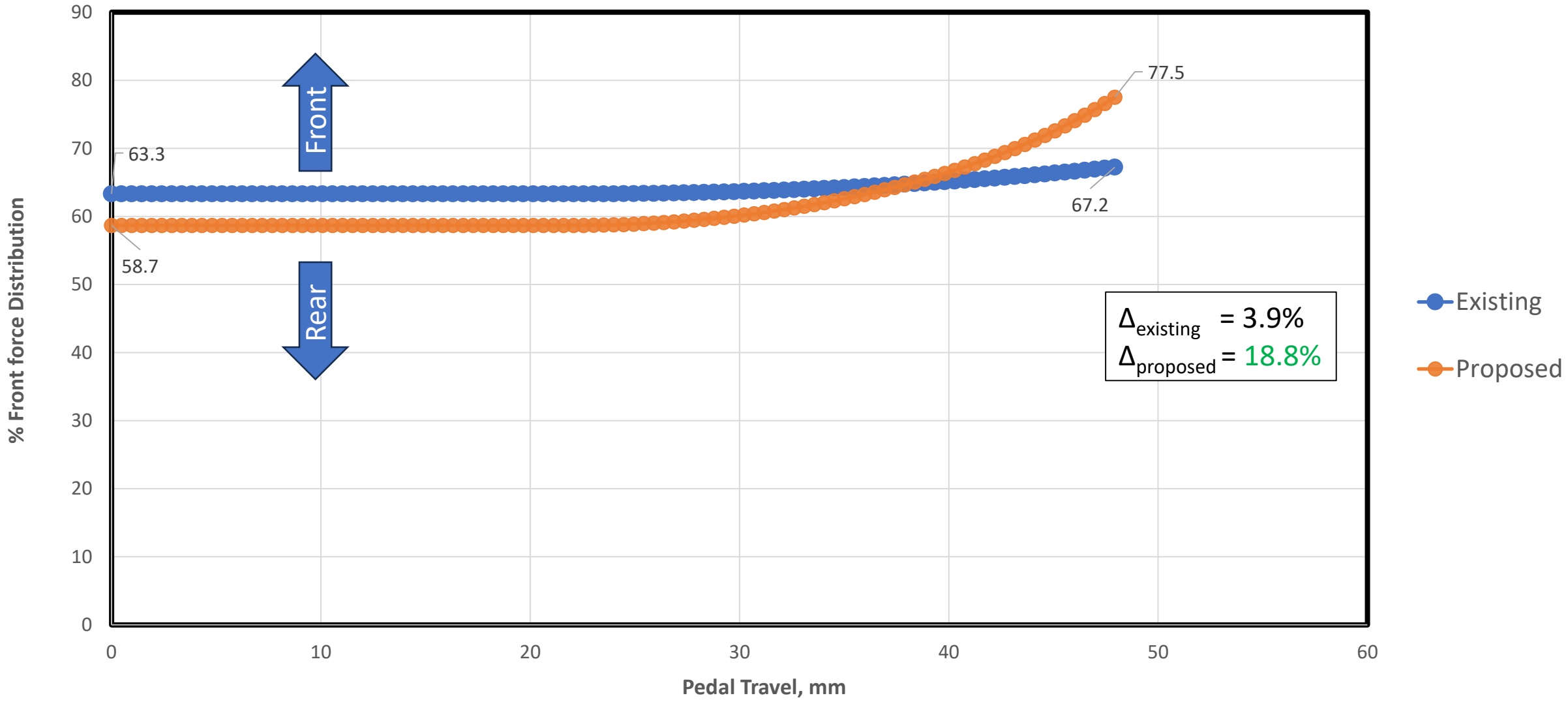
- **A**= Initial angle between brake lever and equalizer link
- **B**= Secondary link initial input length
- **C**= Angle between s.no 9 & 10
- **D**= Brake lever leverage input length
- **E**= Secondary link output length



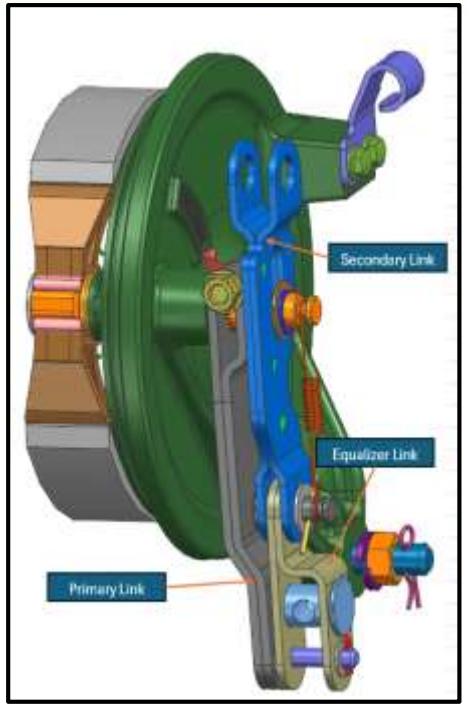


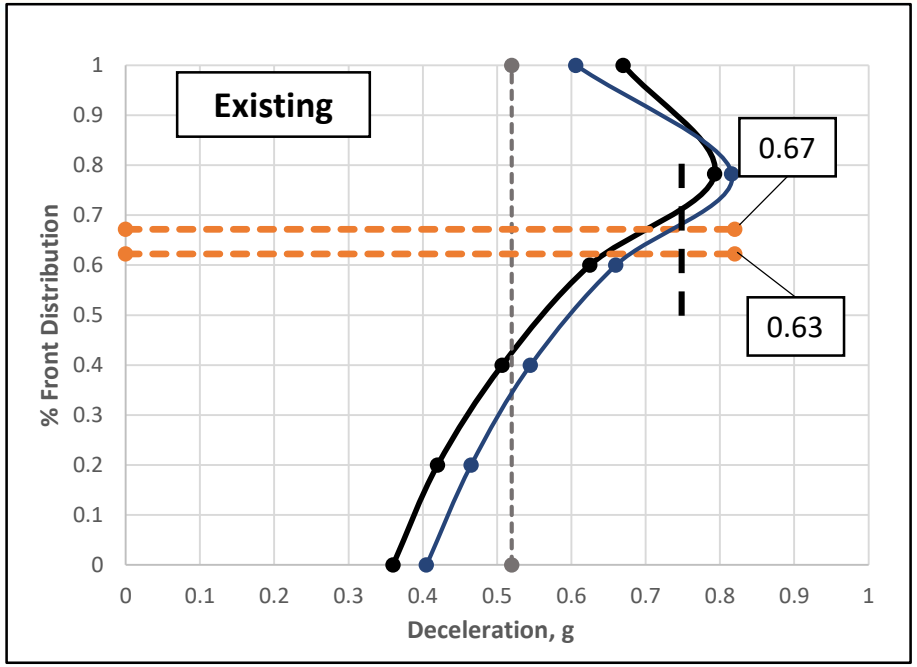


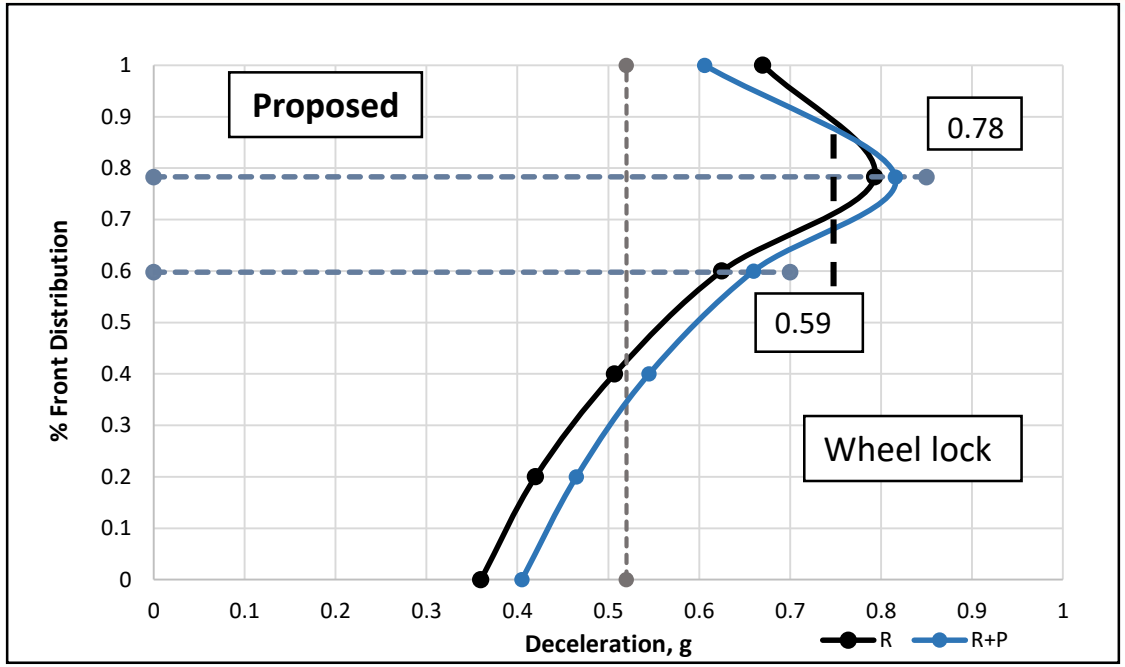
Existing vs Proposed- Force Distribution



Performance Improvement- Deceleration





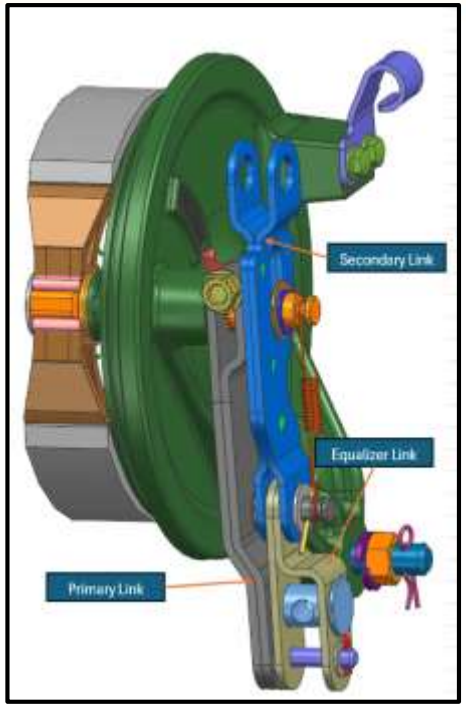


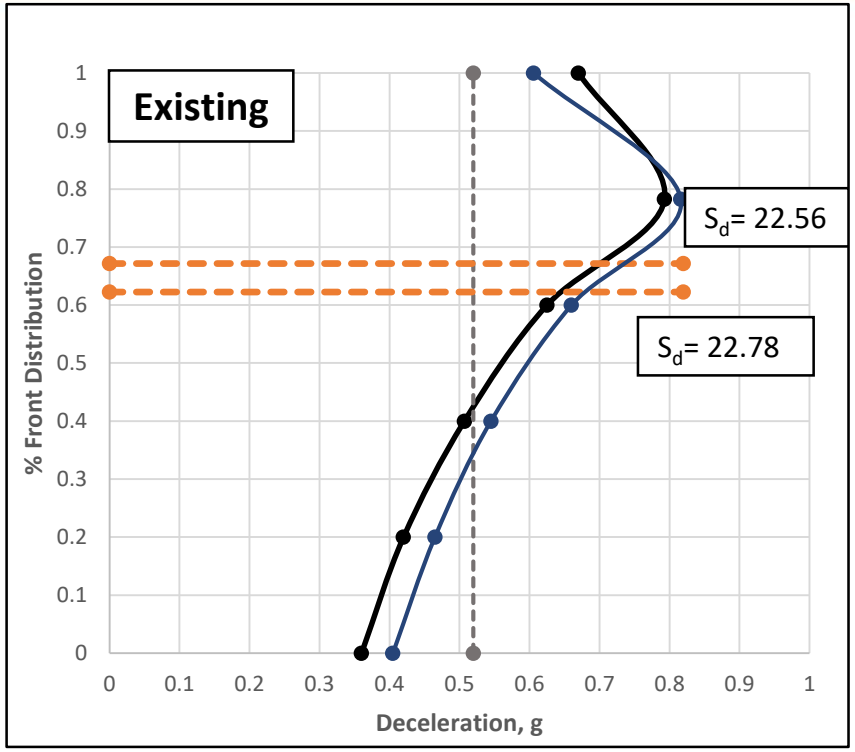
- Current CBS Mechanism meets regulation without *sufficient deceleration buffer.*
- The variable ratio helps in achieving higher decelerations without wheel lock and better stopping distances.

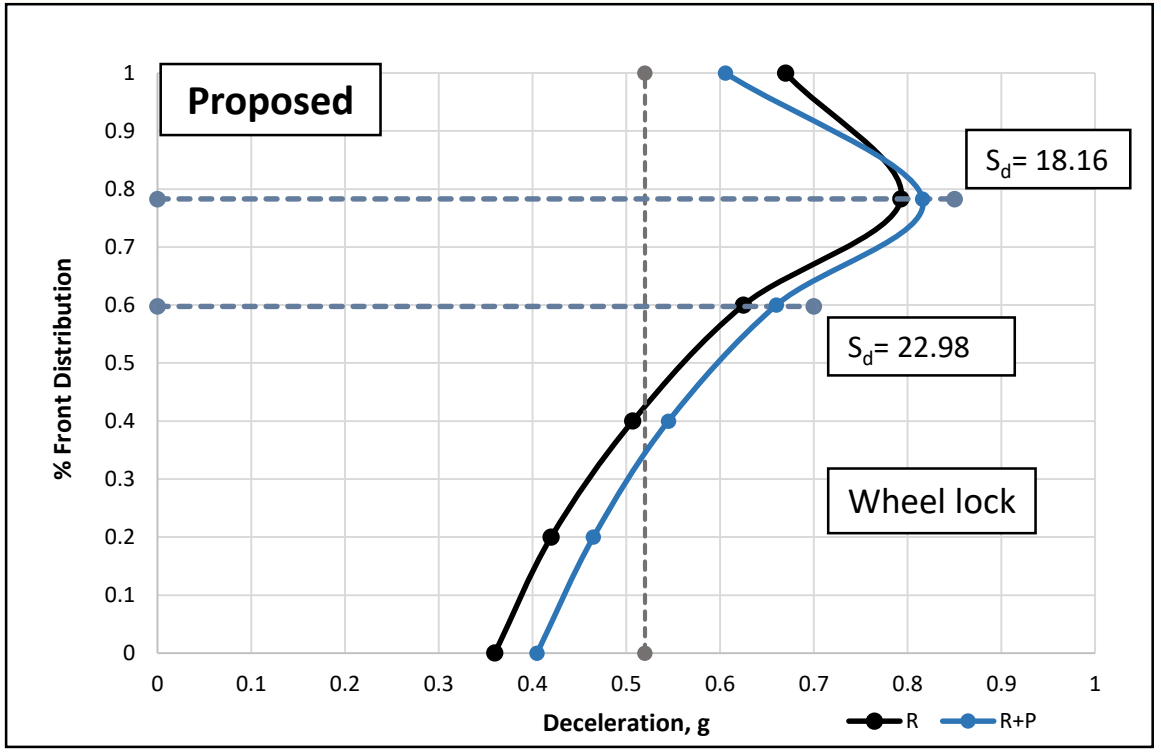
Existing Ratio CBS		
Pedal force, Kg	Deceleration, g	
18	0.42	No Wheel Lock
29	0.68	No Wheel Lock
33	0.78	Wheel Lock

Proposed Variable Ratio CBS		
Pedal force, Kg	Deceleration, g	
18	0.42	No Wheel Lock
29	0.68	No Wheel Lock
33	0.78	No Wheel Lock

Performance Improvement- Stopping Distance







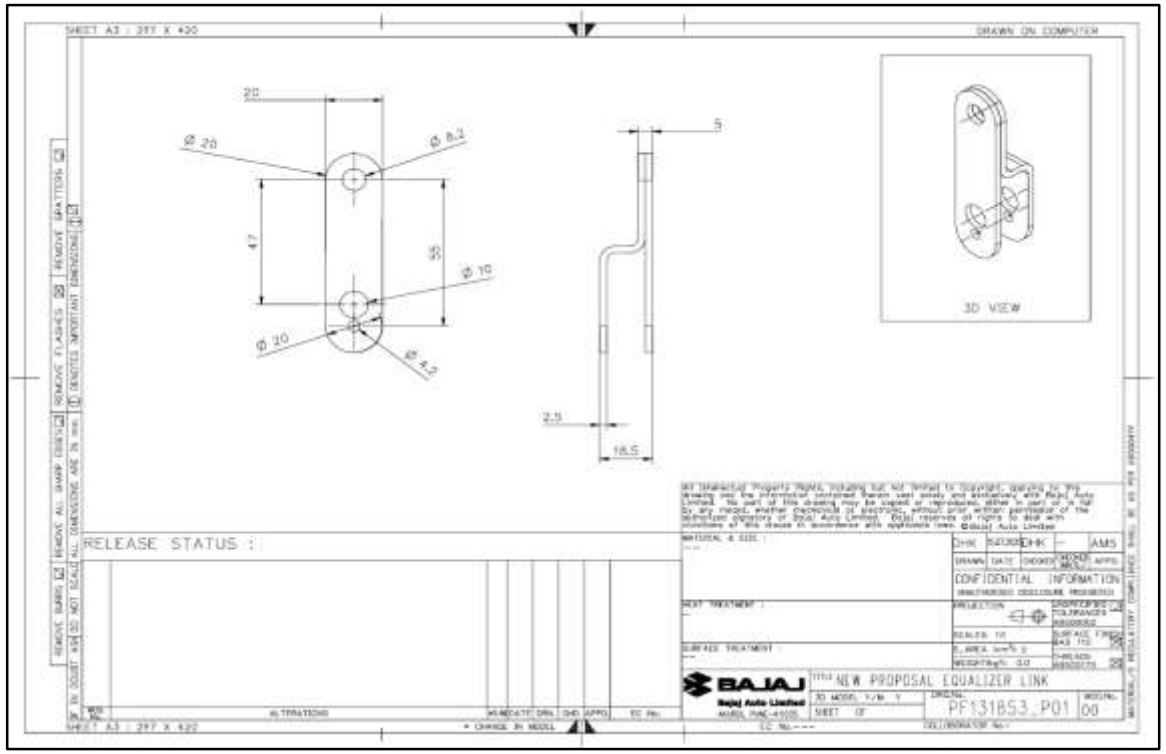
- Current CBS Mechanism meets regulation without *sufficient deceleration buffer.*
- The variable ratio helps in achieving higher decelerations without wheel lock and better stopping distances.

Test Speed
60kmph to 0kmph

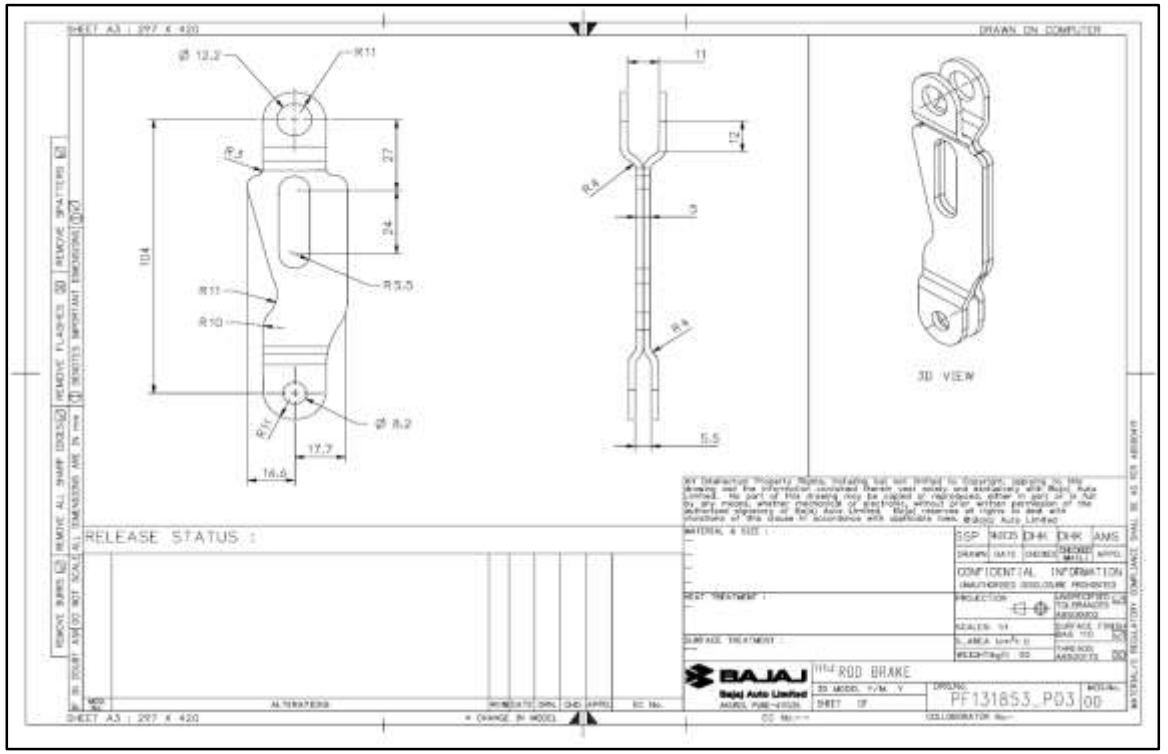
Stopping Distance Comparison		
CBS	Front Ratio	Best Stopping Distance
Existing	0.63-0.67	23 m
Proposed	0.59-0.78	18 m

Proposed Equalizer& Secondary link Configuration- Proto release

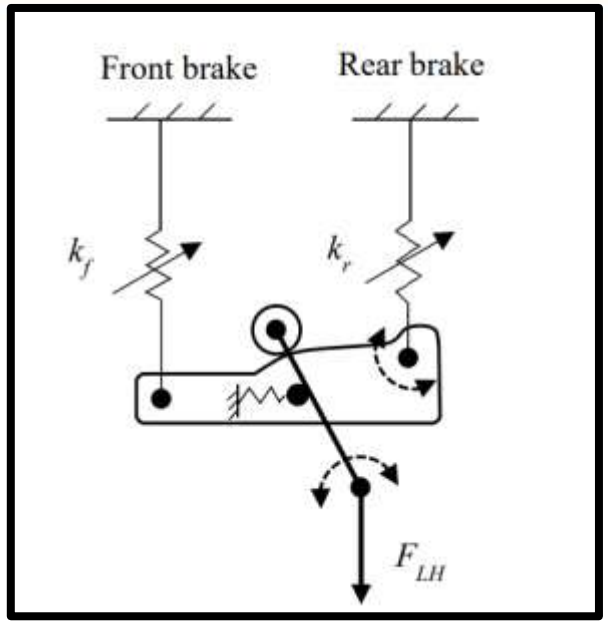
Equalizer Link

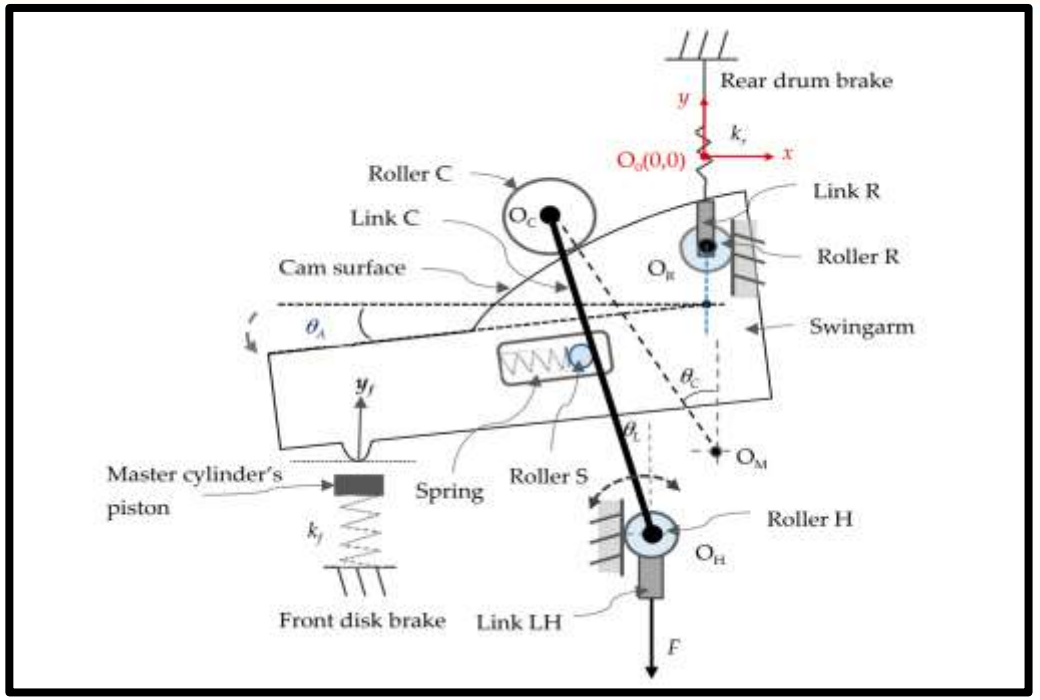


Secondary Link

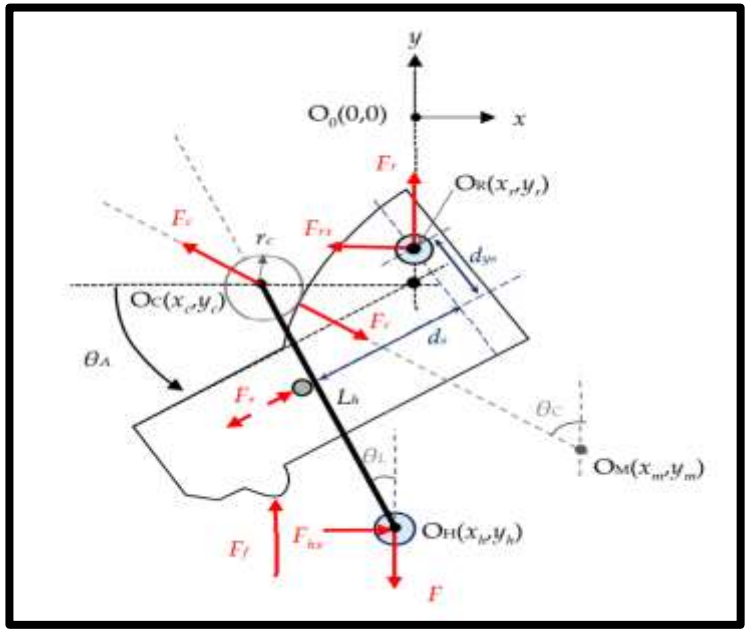


Future Alternative Proposal



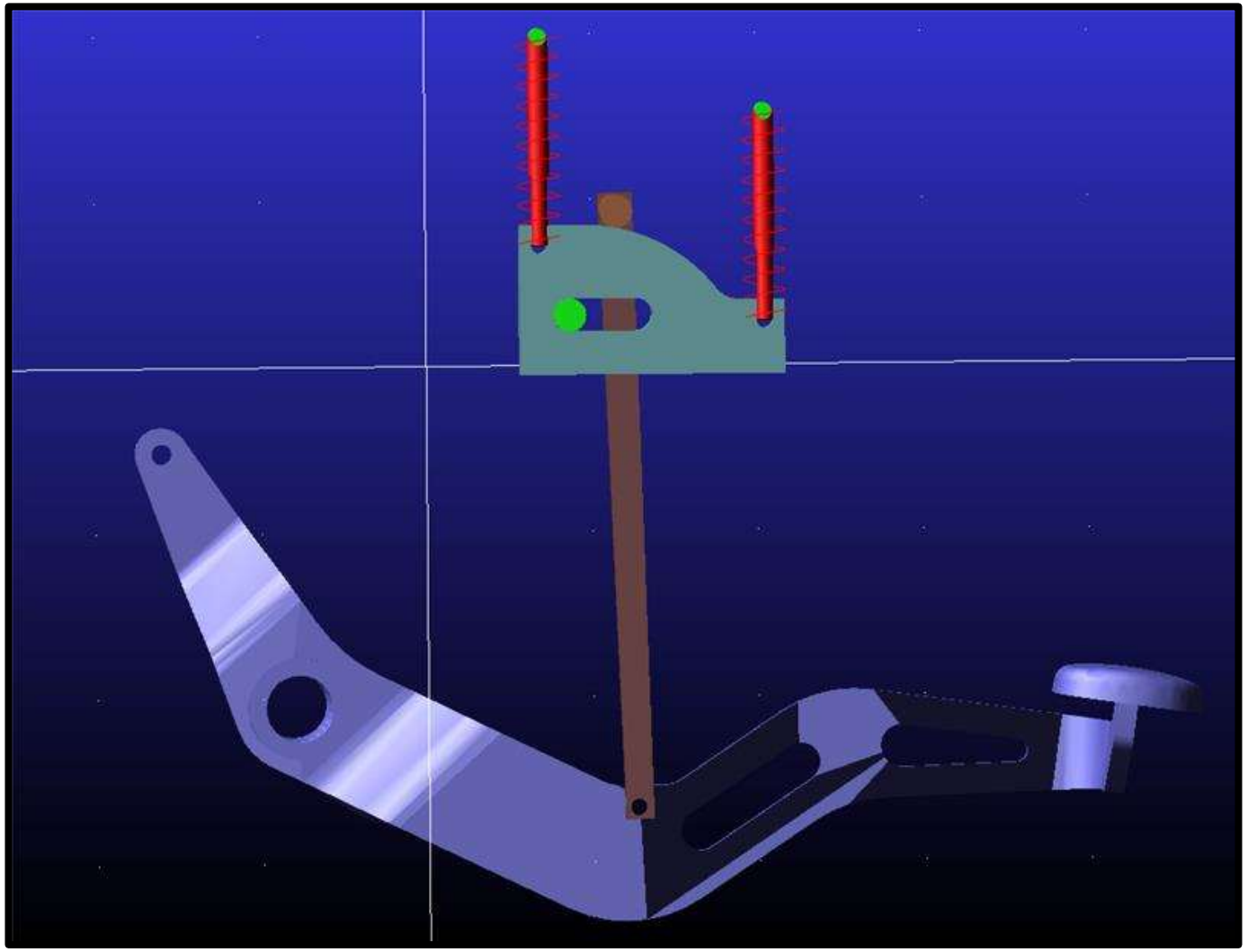


Spring mounted in the slot: primary role is to provide a balancing force to enable the roller C to move along the swingarm's cam surface. **This movement varies the force distribution ratio**



The input force F is from the pedal. A floating swingarm is utilized to distribute the input force to the front and rear brake systems.
The input force originating from the pedal, pulls link C.
The link causes roller C to act upon the swingarm's cam surface.
The input force acting on roller C generates a force component that moves roller C along the cam surface to balance position.

Alternate Proposal





Thank You